

Domains and subtypes

Domains

Domains are used to describe the values accepted in a field. Only values or ranges of values that have been deemed valid can be used as input in a field where a domain is applied. Therefore, domains are an important resource for maintaining data integrity, especially when data is being collected in the field by many workers.

There are two types of domains that can be assigned to a field: coded value domains and range domains.

Coded value domains

Coded value domains specify a valid set of values that can be used in the field. A coded value domain consists of codes that will be saved in the dataset and user-friendly descriptions that correspond to each code, similar to a reference table. For example, an electrical utility provides a map of outages. To maintain consistent information and style on the map, the **Status** field has a domain applied with the accepted status values. The following table is an example of the codes and descriptions that could be applied to the **Status** field:

Code	Description
A	Out of service
B	Dispatched
C	Planned outage

Range domains

Range domains are available for numbers and dates and specify the minimum and maximum values accepted in the field. Data entries outside the range will not be accepted in the field. For example, a public works department must do annual inspections for all the fire hydrants in the city. The normal pressure range for the hydrants is between 50 and 120 psi. A range domain can be created with a minimum value of 50 and a maximum value of 120. The domain can then be applied to the **Pressure** field so only values within that range can be entered. Any hydrants that do not have pressure values in that range will fail the inspection.

Subtypes

Subtypes are a set of features that share the same attributes. Each subtype is defined to provide other fields with the correct attribute information, including domains and default values.

Subtypes are most useful in situations in which the attributes in your data are linked to other attributes. For example, street subtypes could include residential, arterial, and highway. Each of these subtypes could also have an associated default speed limit.